COMSATS University Islamabad, Vehari Campus

Project Proposal

(SCOPE DOCUMENT)

for

FinanceFlow

Version 1.0

By

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SCOPE REVSION HISTORY

No.	Comment	Actions
1	Initial draft submitted to supervisor for review.	Awaiting feedback to incorporate changes in Version 1.1.
2	Supervisor feedback received for Problem and Statement is to long.	Revised the scope and make the problem and solution Statement more short and understandable.
3	Identified need for enhanced offline functionality based on user survey feedback.	Added AsyncStorage integration for robust offline transaction support and updated scope to emphasize offline mode in Version 1.2.
4	Project timeline adjusted to accommodate extended testing for AI-driven categorization accuracy.	Extended testing phase by two weeks and updated Gantt chart in Version 1.4 to reflect new milestones
5	Extended AI testing timeline.	Added two weeks to testing, updated Gantt chart in Version 1.4.

Supervisor Signature:_____

Date:_____

Table of Contents

Abstract 1

1. Introduction 2

2. Problem Statement 2

3. Problem Solution for Proposed System 2

4. Related System Analysis/Literature Review 2

5. Advantages/Benefits of Proposed System 3

6. Scope 3

7. Modules 3

8. System Limitations/Constraints 3

9. Software Process Methodology 4

10. Tools and Technologies 5

11. Project Stakeholders and Roles 5

12. Team Members Individual Tasks/Work Division 6

13. Data Gathering Approach 6

14. Concepts 7

15. Gantt chart 7

16. Mockups 7

17. Conclusion 9

References 9

Plagiarism Report 10

Project Category: (Select all the major domains of proposed project**)**

○ **A-**Desktop Application/Information System ○**B-**Web Application/Web Application based Information System ○**C-** Problem Solving and Artificial Intelligence ○ **D-**Simulation and Modeling ○ **E-** Smartphone Application ○ **F-** Smartphone Game ○ **G-** Networks ○ **H-** Image Processing ○Other (specify category) _____

Abstract

FinanceFlow is a mobile app for personal finance management, enabling users to track transactions, set budgets, manage goals, and receive AI-driven insights and notifications. Built for iOS and Android using React Native and Expo, it addresses gaps in existing apps like Mint and YNAB, which lack advanced notifications and AI categorization. Using Supabase for data sync and Expo for notifications, FinanceFlow offers a user-friendly, secure solution, showcasing software engineering skills for a final year project.

1. Introduction

FinanceFlow is a mobile app for tracking transactions, budgets, and savings goals with AI insights and notifications. Built with React Native and Expo for iOS and Android, it tackles deficiencies in existing budgeting apps, such as limited notifications and goal tracking. Using Supabase for data sync, it provides an intuitive, secure platform, showcasing advanced mobile development for a final year project.

2. Problem Statement

Budgeting apps like Mint and YNAB lack AI-driven transaction categorization, hybrid notifications (real-time and scheduled), and robust goal tracking with progress alerts. High costs and limited multi-currency support reduce accessibility. A free, mobile-only app with these features is needed for effective financial management..

3. Problem Solution for Proposed System

FinanceFlow offers a mobile-only iOS/Android app using React Native and Expo, with Supabase for data sync. It features AI-driven category suggestions, spending analysis, and multi-currency support. Hybrid notifications (immediate via Expo, scheduled via background tasks) and goal tracking with progress alerts enhance usability. The secure, intuitive UI supports efficient financial planning.

4. Related System Analysis / Literature Review

FinanceFlow addresses gaps in mobile budgeting apps lacking its capabilities:

- **Mint**: Tracks transactions but lacks AI categorization, hybrid notifications (no scheduled summaries), and advanced goal tracking. Available at: https://mint.intuit.com/.
- **YNAB**: Offers budgeting but misses AI insights, scheduled notifications, and robust goal management with reminders. Available at: https://www.ynab.com/.
- **PocketGuard**: Provides basic insights but lacks AI categorization, hybrid notifications, and goal tracking. Available at: https://pocketguard.com/.

Related System Analysis: FinanceFlow stands out with AI-driven categorization, immediate/scheduled notifications, and goal management with progress alerts, unlike competitors with basic alerts and limited goal features.

Application Name	Weakness	Proposed Project Solution
Mint	No AI categorization, weak notifications, no goal alerts.	AI category suggestions, hybrid notifications, goal tracking.
YNAB	No AI, basic alerts, weak goal features.	AI insights, advanced notifications, goal progress alerts.
PocketGuard	No Goals, no hybrid alerts	AI categorization, comprehensive notifications, goal management.

5. Advantages/Benefits of Proposed System

- **Mobile Accessibility**: Runs on iOS and Android.
- **AI Insights**: Category detection, spending patterns, top categories.
- **Notifications**: Immediate (e.g., overspending) and scheduled (e.g., weekly summaries).
- **Efficiency**: React Native and Expo for streamlined development.
- **Security**: Supabase authentication and Row Level Security.
- **UI**: Intuitive with gradient backgrounds and modals.
- **Goals**: Savings targets with progress tracking and alerts.

6. Scope

FinanceFlow is a mobile-only app for personal finance, built with React Native and Expo for iOS and Android. It supports transaction tracking, budgeting, goal setting, and AI-driven insights with hybrid notifications. Key features include:

- **Authentication**: Supabase-based login, profile, and currency settings.
- **Transactions**: CRUD with AI category suggestions (e.g., "Starbucks" → Food) and sortable history.
- **Budgets**: Category limits with real-time tracking and alerts.
- **Goals**: Savings targets with progress, contributions, and reminders.
- **AI Insights**: Spending patterns and top categories with multi-currency support.
- **Notifications**: Immediate (Expo) and scheduled (background tasks) alerts.
- **Support Screens**: Privacy, terms, help, and about.

7. Modules

FinanceFlow is divided into seven modules for modularity:

- **Module 1: Authentication & User Management**

  Provides secure user registration and login via Supabase, ensuring robust authentication. Users can manage profiles, including personal details and preferences, with multi-currency selection. React Context maintains global state for currency formatting across the app.
- **Module 2: Transaction Management**

  Enables CRUD operations for income and expense transactions with input validation. AI-driven category suggestions (e.g., "Starbucks" → Food) enhance user efficiency. A sortable and searchable transaction history offers quick access to financial records.
- **Module 3: Budget Management**

  Allows users to set monthly category limits (e.g., $200 for Dining) with real-time tracking. Progress bars visualize spending, and alerts notify users of overspending risks. A toggle system enables easy activation or deactivation of budget constraints.
- **Module 4: Goal Management**

  Supports setting savings goals with due dates and progress tracking via visual indicators. Users can contribute funds to goals, triggering celebration notifications upon milestones. Time-based reminders ensure users stay on track with their financial objectives.
- **Module 5: AI-Powered Insights**

  Delivers spending pattern analysis and top category identification using lightweight AI. Automatic category detection assigns labels based on transaction descriptions. Multi-currency support ensures accurate insights tailored to user preferences.
- **Module 6: Notification System**

  Manages immediate notifications (e.g., overspending alerts) via Expo and scheduled ones (e.g., weekly summaries) using background tasks. A dedicated notifications screen displays all alerts with options to mark as read. Background processing ensures reliable delivery of time-based updates.
- **Module 7: Support Screens**

  Includes screens for privacy policy and terms of service to ensure transparency. A help section provides FAQs and support contact for user assistance. The about screen details app information and developer credits for user reference.

8. System Limitations/Constraints

- No bank integration; manual inputs only.
- Lightweight AI, no extensive ML training.
- Mobile-only features (notifications, biometrics).
- Supabase free tier limits scalability.
- Partial offline sync; full sync needs internet.

9. Software Process Methodology

The project adopted **Agile Methodology** with bi-weekly sprints to ensure flexibility and iterative development. Each sprint involved planning, coding, testing, and review to align with the 6-12 month timeline. This approach enabled modular development with React Native and Expo, facilitating rapid prototyping and team collaboration.

10. Tools and Technologies

Category	Tools/Technology	Version	Rationale
Tools	Visual Studio Code	Latest (2025)	IDE for coding and debugging
	Android Studio	Latest (2025)	Emulator for Android testing
	Xcode	15.x	Emulator for iOS testing
	MS Word	2021	Documentation
	MS PowerPoint	2021	Presentation
	Figma	125.4.9	Mockup creation
Technologies	React Native	0.74.x	Cross-platform development
	JavaScript	ES2023	Programming language
	Expo SDK	Latest (2025)	Development and notifications
	TypeScript	Latest	Cross-platform desktop packaging
	Supabase	Latest (2025)	Authentication and cloud storage
	expo-notifications	Latest	Push/local notifications
	Python	3.11.x	Lightweight AI for transaction categorization and spending analysis
	Node.js	20.x	Backend and build tools

11. Project Stakeholders and Roles

Project Sponsor	COMSATS University, Islamabad

Asadullah	Developer: Implements User Account Management, Transaction Management, and Intelligent Analysis modules.
Muhammad Taimoor	Developer: Budgets, Goals, Notifications, Support Screens
Komal Hassan	Project Supervisor: Provides guidance, reviews progress, and ensures academic compliance.
Final Project Committee	Year Evaluates project deliverables, scope, and presentation for grading.

12. Team Members Individual Tasks/Work Division

Student Name	Student Registration Number	Responsibility/ Modules
Asadullah	FA22-BSE-037	User Account Management, Transaction Management, Intelligent Analysis (Modules 1, 2, 5). Handles Supabase integration, AI logic (spending trends, auto-categorization, predictive alerts, savings recommendations), and mobile testing.
Muhammad Taimoor	FA22-BSE-072	**Budgets, Goals, Notifications, Support Screens (Modules 3, 4, 6, 7)**: Designs budget and goal management with real-time tracking and alerts. Builds hybrid notification system using Expo and background tasks, plus support screens. Creates intuitive UI with gradient backgrounds and modals.

13. Data Gathering Approach

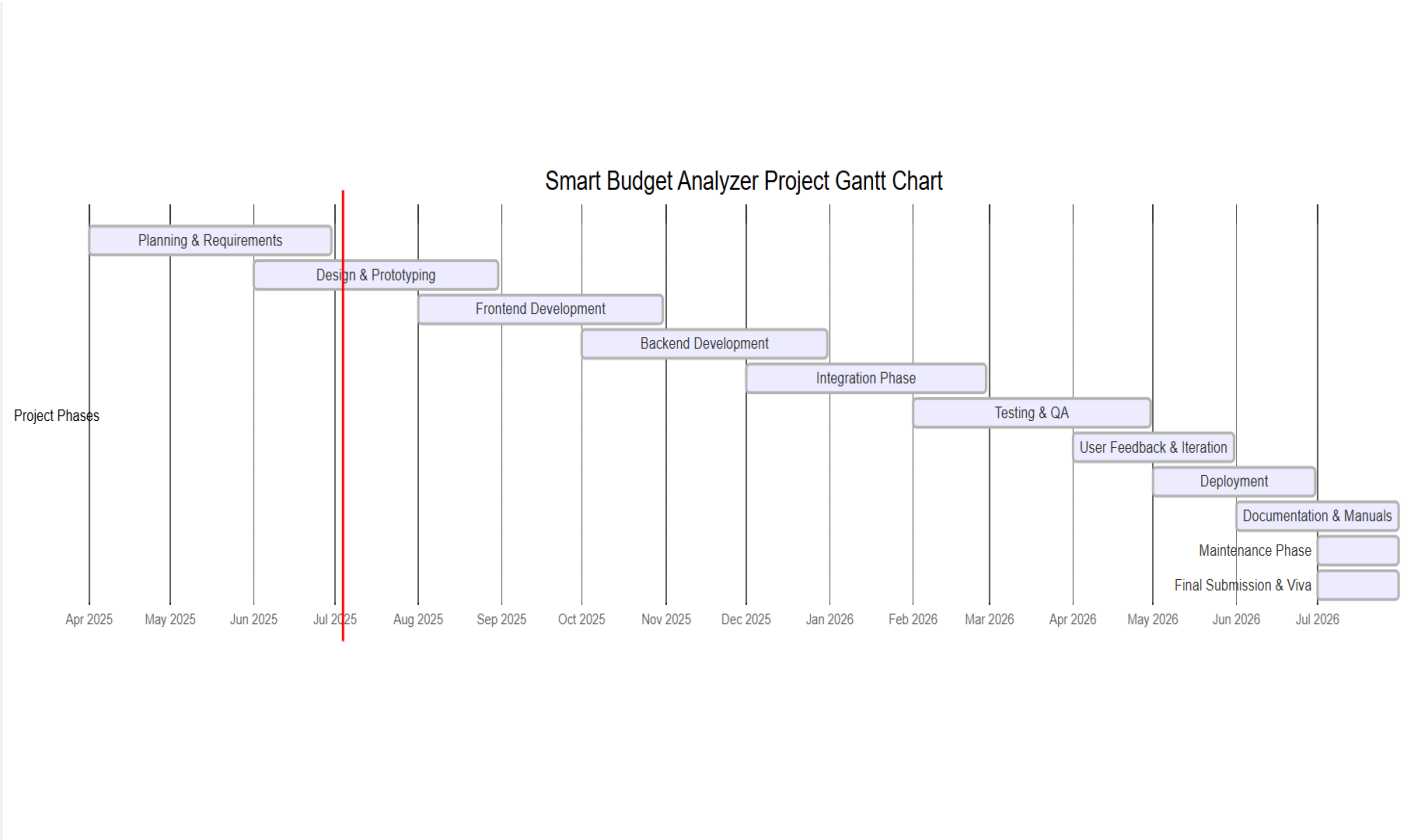
Surveys and interviews with 20 users identified needs for AI, notifications, and goals. App analysis (Mint, YNAB) highlighted gaps. Supervisor feedback ensured academic alignment.

Form link:
https://docs.google.com/forms/d/e/1FAIpQLSd43c0IWF9fTvlLig3oTRxkwfhcvEAjT9rFoKbbfeyeWBuA6A/viewform?usp=dialog

14. Concepts

- **Mobile Development**: React Native and Expo for iOS/Android.
- **Cloud Sync**: Supabase for real-time data management.
- **AI**: Lightweight categorization and insights.
- **Notifications**: Hybrid system with Expo and background tasks.

15. Gantt chart



16. Mockups

Four mockups will be designed using Figma to represent major modules:

1) Dashboard:

Main financial overview and navigation hub:-

User Header : Personalized greeting with notifications and settings access

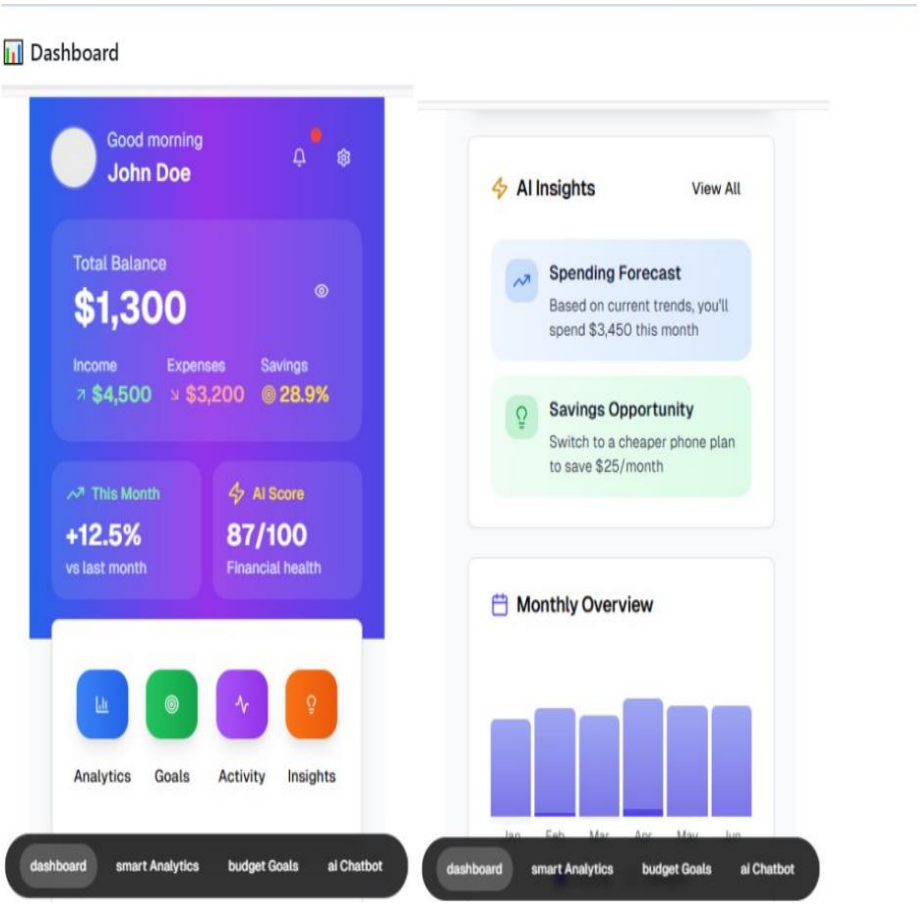
Balance Card : Total balance display with income/expense breakdown and privacy toggle

Quick Stats : Monthly trend percentage and AI financial health score

Action Grid : Four-button navigation to Analytics, Goals, Activity, and Insights

AI Preview : Top 2 personalized financial recommendations with "View All" option

Monthly Chart : 6-month spending visualization comparing actual vs budget amounts



2) Smart_Analysis:

Detailed financial data analysis and insights:

Analytics Header : Period summary cards with share/download export options

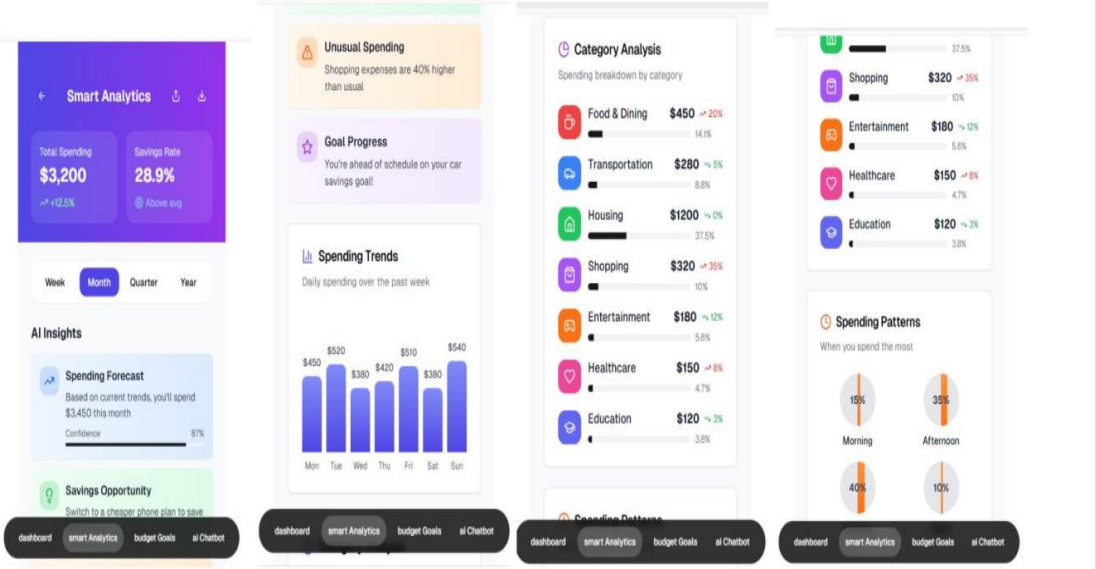
Period Selector : Week/Month/Quarter/Year time frame selection buttons

AI Insights : Predictive forecasts, savings opportunities, alerts, and achievements

Trends Chart : Interactive 7-day spending breakdown with daily amounts

Category Analysis : Spending breakdown by category with trend indicators and percentages

Spending Patterns : Time-based analysis showing when you spend most (morning/afternoon/evening/night)



3) Budget_goals:

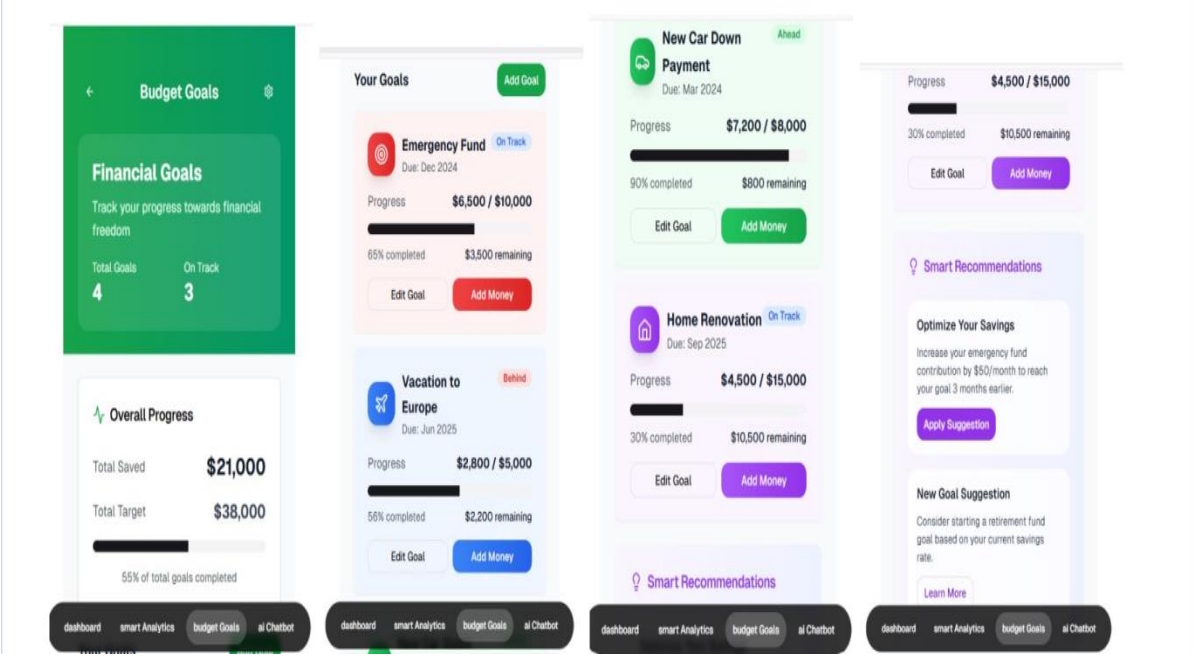
Financial goal tracking and management :-

Goals Header : Total goals count and on-track performance summary

Progress Overview : Combined savings total with master completion percentage

Goal Cards : Individual goal tracking with progress bars, status badges, and action buttons

Smart Recommendations : AI-powered optimization tips and new goal suggestions



17. Conclusion

FinanceFlow is a mobile-only app for personal finance, offering transaction tracking, budgeting, goals, AI insights, and hybrid notifications. Built with React Native, Expo, and Supabase, it addresses gaps in existing apps, delivering a secure, intuitive solution for a final year project.

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[1] React Native Documentation, **"React Native: A framework for building native apps using React,"** [Online]. Available: https://reactnative.dev/. [Accessed: May 7, 2025].

[2] Supabase Documentation, **"Supabase: The Open Source Firebase Alternative,"** [Online]. Available: https://supabase.com/docs. [Accessed: May 7, 2025].

[3] TensorFlow.js, **"Machine Learning in JavaScript,"** [Online]. Available: https://www.tensorflow.org/js. [Accessed: May 7, 2025].

[4] M. Smith, **"Personal Finance Apps: A Comparative Study,"** Journal of Financial Technology, vol. 12, no. 3, pp. 45-60, 2023.

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